FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-05 - Start Here

Exact study order, setup boundary and learner-machine runtime contract.

Artifact	C3-05-START
Status	2026 - LEARNER EDITION
Phase	C3-05 Distributed Processing with Spark | 16 lessons | 46-60 focused hours
Prerequisite	C3-04 Gate PASS; C2B SQL/Python/modeling foundations assumed, distributed execution taught here.
Current lane	Apache Spark / PySpark 4.2.0 | Python 3.10+ | Java 17+ | learner local[2]
Brand	Charcoal #1F2937 | Teal #14B8A6 | Soft Blue #3B82F6 | AMOLED #000

House teaching rule: mental picture -> tiny worked example -> guided real Spark workflow -> expected result -> why it worked -> break/fix -> independent changed task -> validation -> physical-plan/production reasoning -> evidence. A green Spark job is not mastery if grain, quality, plan or reproducibility is wrong.

What you will build

By the end of C3-05 you can build a correct PySpark transformation, prove data quality and grain, inspect distributed execution, diagnose partitions/skew, test it, and hand it off as a repeatable job.

First-use setup - only now

- Use WSL2/Linux workspace from C3-00. Keep Docker, Airflow and Kafka closed unless an integration exercise specifically requires them.
- Verify: python3 --version (3.10+). Verify: java -version (Java 17+). Verify JAVA_HOME if Java is not found.
- Create a dedicated C3-05 venv: python3 -m venv .venv && source .venv/bin/activate.
- Install from starter requirements: python -m pip install -r requirements.txt. The current learning pin is pyspark==4.2.0 and pytest==9.0.2.
- Verify: python -c "import pyspark; print(pyspark.__version__)" should report 4.2.0. Start local sessions with local[2]; do not allocate every CPU/memory resource on an 8 GB laptop.

If installation/runtime is blocked, continue the book and pure-Python/source-control tasks, mark literal Spark proof PENDING, and return to the runtime lane later. Never invent Spark UI, plan or Parquet evidence.

Exact study route

- Open C3-060 lesson page shown by Mentor route.
- Follow the real starter project once.
- Attempt matching C3-061 Pxx and save genuine evidence.
- Only then open C3-062 Rxx; close it and do the changed retry.
- After L16 mastery, attempt C3-063 CourierSpark Mini-Lab.
- Then attempt C3-064A MeterSpark Gate A.
- If repair is needed, use targeted cards + changed micro-retest. Gate B/C remain sealed until explicit full-retest assignment.

Laptop profile

SPARK profile: WSL2 + Java + PySpark local[2] + editor/browser. Keep Airflow/Kafka/Docker closed. Tiny local shuffle partition values are teaching scaffolds; production values must be decided from workload evidence.